Human Ancestors: *Paranthropus robustus*



Habitat and Behaviour

Fossil Record

Anatomical Features

Paranthropus – Overview

- Lived 1.8-1.2 million years ago in southern Africa
- Likely several species including P. boisei but now an extinct line
- Likely branched off evolutionary tree from Au. afarensis – Au. africanus and other Homo species another branch
- Formerly known as “robust Australopithecines” due to skeletal differences with other “gracile Australopithecines”
- Spent time in trees and on ground
- Mostly vegetarian – roots and seeds, but also fruit
- “Nutcracker Man” due to size of teeth and strong chewing muscles



Learning Aim: Describe the main features of Paranthropus

Paranthropus – Fossil Record

- First fossil found in 1938
  - Robert Bloom
  - Jaw fragment and molar
  - Different skeletal structure to other Australo's
  - Para = alongside; anthropus = man
- More fossils found in 1955 in Olduvai Gorge, Tanzania
- Mary Leakey (1959)
  - "Zinj" skull
  - P. boisei classified
- Eventually three separate Para species identified





Learning Aim: Describe the fossil record for Paranthropus

Paranthropus – Anatomical Features - Dentition

Australopithecus		Human
• Canines shorter than chimp, larger than human, non projecting • Small diastema present • Dental arcade becoming parabolic • Teeth much larger than human – indicates plant based diet.		• Canines small • No diastema • Parabolic dental arcade • Teeth small – indicates omnivorous diet and less prognathism.

Learning Aim: Describe dentition of Paranthropus and compare to Australopithcines and humans

Australopithecines – Anatomical Features – Skull

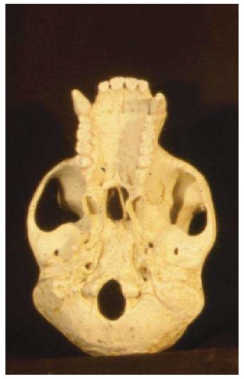
- Facial profile
  - Lower forehead, more projecting jaw than more modern hominins
  - Brain size 480cc, closer to modern gorilla than human
  - Relative brain size larger though: Australopithecus smaller than gorilla

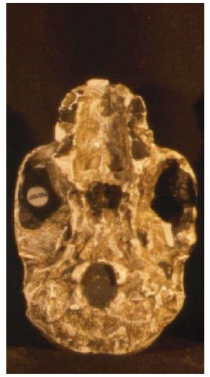


- Brain size and anatomy
  - Brain size 480cc, closer to modern gorilla than human
  - Relative brain size larger though: Australopithecus is smaller than gorilla
  - Endocasts show brain more similar to humans in structure
  - Foramen magnum more forward than in apes
  - Skull more rounded in back than in apes







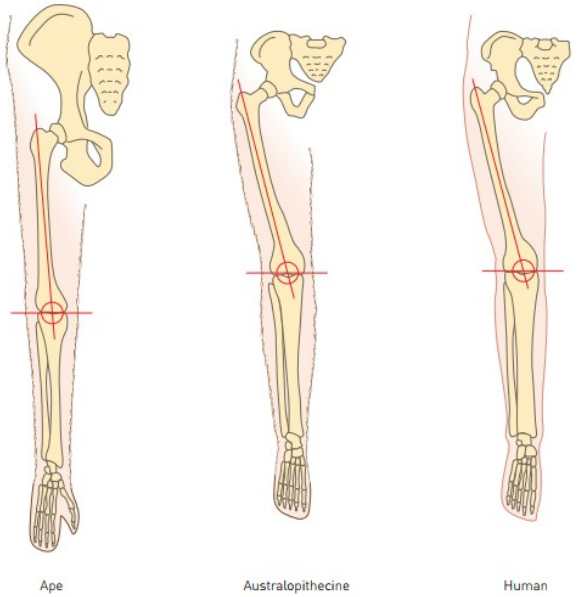


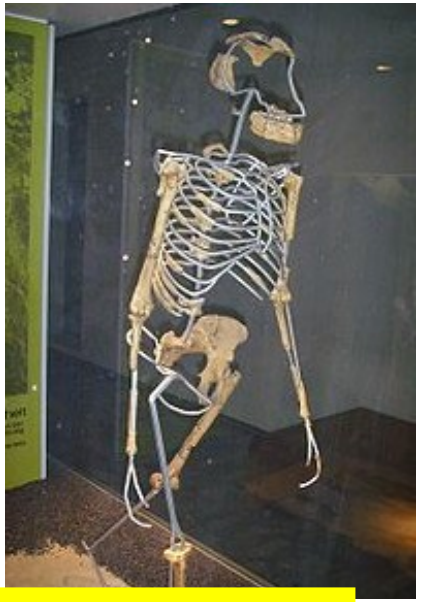


Learning Aim: Describe the skull of Australopithecines and compare to modern apes and humans

Australopithecines - Anatomical Features - Limb bones

- Footprints show bipedal gait, although not quite the same as modern humans
- Pelvis and carrying angle more similar to modern humans than to apes
- Limb bones
  - Suggest bipedal locomotion
  - Pelvic and foot bones typically hominin
  - Non opposable, strongly built big toe
  - Thumb less mobile than modern humans, but more so than modern apes
  - Vertebral column – typically hominin curves
  - Foramen magnum – at base of skull.





Learning Aim: Describe the femur and pelvis of Australopithecines and compare to modern apes and humans

Evolutionary Progression in Australopithecines

- More than one species of Australopithecine.
- *Australopithecus afarensis* – earlier
- *Australopithecus africanus* – later
- Can see evolutionary transition between the two species:

TABLE 13.3 A comparison of *Australopithecus afarensis* and *Australopithecus africanus*

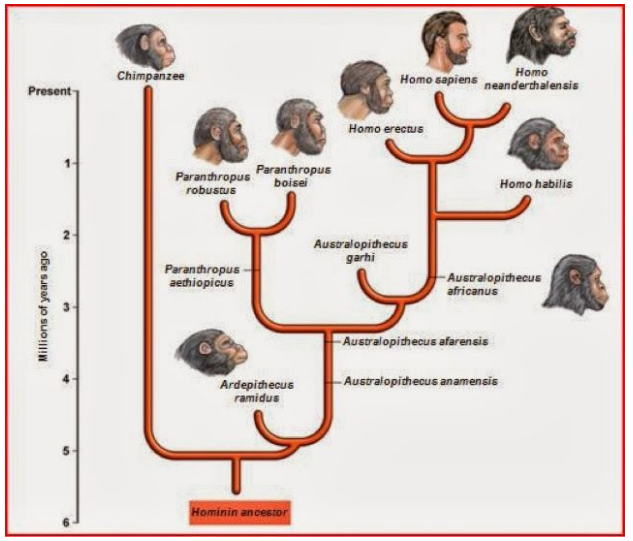
	Australopithecus afarensis	*Australopithecus africanus*
Time of existence	3.9 and 2.8 million years ago	3.2 to 2 million years ago
Location	East Africa	Southern Africa
Height	Female: 105–110 cm Males: 150 cm	Female: 110 cm Males: 135 cm
Brain	430 cm^3	480 cm^3
Skull	Low, sloping forehead Prominent brow ridges Short sagittal crest in males	Slightly arched forehead Smaller brow ridge
Teeth and jaw	Prognathic jaw Small canine teeth (but larger than *A. africanus*) Diastema present	Prognathic jaw Shorter and smaller incisors and canines Large molar and premolars No diastema
Limbs	Big toe not opposable Long arms, although shorter than the legs Long curved fingers and toes Features for bipedalism	Big toe not opposable Long arms, although shorter than the legs Some curvature of the finger and toe bones Features for bipedalism
Pelvis	Short and wide pelvis	Short and wide pelvis, less rounded than in modern humans

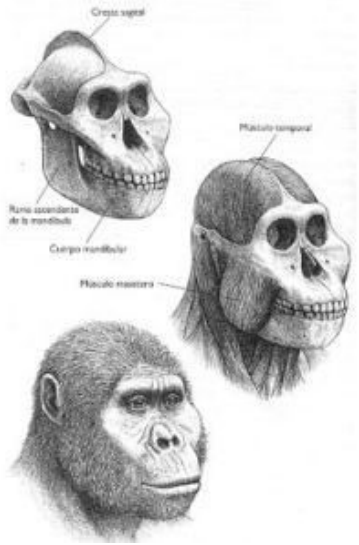
Learning Aim: Discuss features that demonstrate evolutionary progession within the Australopithecus genus.

Paranthropus robustus

- Closely related to *Australopithecus*
- *Paranthropus* used to be classified as *Australopithecus*.
- Different branch of the evolutionary tree.

Australopithecus	*Paranthropus robustus*
120-140cm tall	Up to 170cm tall
25-35 kg	Up to 70 kg
Proportional molars	Larger, broader molars
Smaller jaw muscles	Large jaw muscles and attachments, and sagittal crest
Australopithecus africanus *Australopithecus afarensis* Used to be classified as "gracile" Australopithecus	*Paranthropus robustus* used to be classified as "robust" Australopithecus











Learning Aim: Describe differences between Australopithecus and the closely related Paranthropus.

Australopithecus



Relationship to humans:

- Direct ancestor to humans but also to other extinct hominids, some on other "branches"

Time Period:

- 3-4 mya (Pliocene)
- Several species

Distribution:

- East Africa



Climate:

- Becoming drier:
- Forests becoming grass-land.
- Selection pressure for bi-pedalism and adaptability.

Fossil Record:

Laetoli Footprints

- 3.6 mya
- Discovered 1978—Leakey and team
- Evidence of bipedalism
  - * Aligned big toe
  - * Heel strike, toe off motion



"Lucy" - partial skeleton

- 3.2 mya
- Discovered 1974
- Partial skeleton shows evidence of lifestyle
  - * Strong chest and upper arm—climbing
  - * Short, broad pelvis—bipedal
  - * Limb carrying angle—bipedal
  - * Flat nose, prognathic jaw - ape-like





"Taung Child" - skull with endocast

- 2.8 mya
- Discovered 1924—Raymond Dart
- Skull and endocast:
  - * Foramen magnum beneath skull—upright stance
  - * Endocast shows apelike brain but slightly larger and more complex



Cultural Evolution:

Tool Use

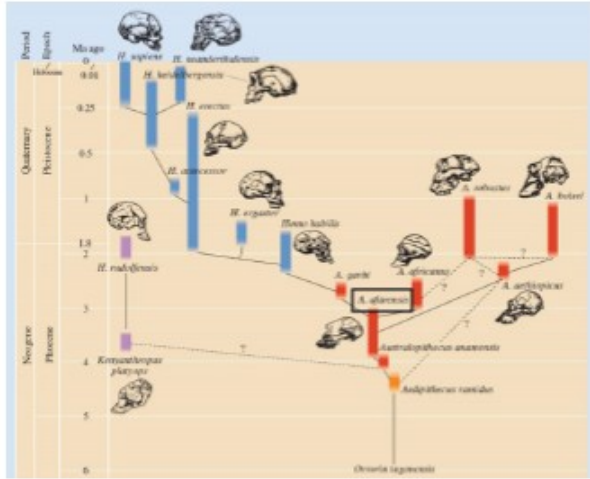
- Used rocks and sticks in foraging eg to "fish" for insects or break nuts—much like chimps today. MAY have crafted primitive stone tools.

Fire Use

- No evidence of fire use

Language, Art, Abstract Thought

- No evidence of complex Language, art or abstract thought



Lifestyle:

- In trees and on ground
- Bipedal gait
- Foraging and scavenging
- Family groups

Diet and Dentition:

- Mostly fruit, leaves, vegetation
- Teeth intermediate between apes and humans:
  - * Rounder jaw than apes
  - * Smaller diastema than apes
  - * Smaller canines than apes

